

Infiltration pattern in a regolith–fractured bedrock profile: field observation of a dye stain pattern

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Abstract:

We examined the infiltration pattern of water in a regolith–bedrock profile consisting of two overburdens (OB1 and OB2), a buried rice paddy soil (PS), two texturally distinctive weathered materials (WM1 and WM2) and a fractured sedimentary rock (BR), using a Brilliant Blue FCF dye tracer. A black-coloured coating in conducting fractures in WM1, WM2 and BR was analysed by X-ray diffraction and scanning electron microscopy. The dye tracer penetrated to greater than 2 m depth in the profile. The macropore flow and saturated interflow were the major infiltration patterns in the profile. Macropore flow and saturated interflow were observed along fractures in WM1, WM2 and BR and at the dipping interfaces of PS–WM1, PS–WM2 and PS–BR respectively. Heterogeneous matrix flow occurred in upper overburden (OB1) and PS. Compared with OB1, the coarser textured OB2 acted as a physical barrier for vertical flow of water. The PS with low bulk density and many fine roots was another major conducting route of water in the profile. Manganese oxide and iron oxide were positively identified in the black coating material and had low crystallinity and high surface area, indicating their high reactivity with conducting contaminants. Copyright © 2006 John Wiley & Sons, Ltd.

KEY WORDS Brilliant Blue dye; layer interface; macropore flow; manganese and iron oxides; saturated interflow

INTRODUCTION

Subsurface water movement is of widespread interest for weathering and soil morphology, for groundwater recharge, for surface water and groundwater interaction and for contaminant transportation and reaction. Homogeneous matrix flow of water and solute can be accurately estimated using numerical models based on a Darcian assumption and field measurement (Warrick *et al.*, 1997). However, a precise estimation of solute transport by a preferential flow in the subsurface is difficult because of the heterogeneity of natural porous media. The preferential subsurface flow occurs by a number of mechanisms, such as faster conduction through macropores (Beven, 1981; Bouma, 1981), instability of the wetting front (Hill and Parlange, 1972) and lateral flow along a sloping layer interface (Walter *et al.*, 2000).

Macropore flow involves transport through noncapillary large pores (Beven and Germann, 1982). Buttle and Leigh (1997) showed that larger macropores serve as more efficient conduits for the preferential flow of water than smaller macropores. The soil with the higher density of macropores had the greater drainage area (Weiler and Naef, 2003). Continuous macropores exhibited faster and larger water conduction than isolated macropores (Allaire-Leung *et al.*, 2000a,b). Allaire-Leung *et al.*, (2000a,b) also showed that macropores with a greater tortuosity had a greater retardation time and a greater infiltration of conducting water into the surrounding matrix.

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Hill and Parlange (1972) demonstrated first that 'fingering' could occur in a fine-over-coarse-textured soil. When the conductivity of the coarse-textured sublayer is greater than the rate of transmission through the fine-textured upper layer, the flow velocity increases across the interlayer plane. The increased velocity leads the fingering (Baker and Hillel, 1990). Fingering was also reported in water-repellent soils (Dekker and Ritsema, 1994). The water repellency of a soil depends on the content of organic matter and water (Bisdorn *et al.*, 1993; Dekker and Ritsema, 1994). Fingering also occurs when the air pressure increases ahead of the infiltration front (Hill and Parlange, 1972).

There are two types of lateral flow along a sloping layer interface: saturated interflow (subsurface storm flow or through flow) and funnel flow (Walter *et al.*, 2000). The saturated interflow occurs in the upper soil region underlain by a hydraulically restrictive layer, such as massive bedrock and fragipan (Hewlett and Hibbert, 1963; Stagnitti *et al.*, 1986). The vertical movement of water in the upper layer may pause due to the lower permeability of the underlain layer and then water may be accumulated at the interface. The accumulated water flows along the sloping interface (Kung, 1990). The funnel flow occurs at the sloping interface of the upper fine layer and lower coarse layer in soils. A capillary barrier forms at the sloping interface and it leads the funnel flow. When the matrix potential at the textural interface is too low to move into the lower coarse layer, the capillary barrier restricts the vertical movement of water, thus forcing downward movement along the interface. Three flow regimes (capillary diversion, partial breakthrough and complete breakthrough) develop along the sloping interface if the interface is sufficiently long (Walter *et al.*, 2000).

Dye tracing using a nonfluorescent food dye Brilliant Blue FCF ($C_{37}H_{34}N_2Na_2O_9S_3$) has been employed to visualize flow paths and to mimic the transport behaviour of solutes in the subsurface (Flury and Jury, 1995). Understanding migration and distribution of contaminants in the subsurface is essential for the reduction and prevention of contamination of soil and groundwater and for the investigation and remediation of a contaminated site. The objective of this study was to examine the flow paths in a profile of regolith-bedrock located in an industrial site in South Korea and to characterize minerals, the potential reactants with solute and contaminant in the flow paths.

MATERIALS AND METHOD

Study site

The study site is located at 20 km northeast of Daegu, South Korea (Figure 1). The site was originally a rice paddy and was recently developed for an industrial site. A schematic presentation of the profile is shown in Figure 2. The study site had a profile consisting of a sedimentary bedrock (BR), two layers of texturally distinctive weathered material of sedimentary rock (WM1 and WM2), a buried rice paddy soil (PS), and two layers of overburden (OB1 and OB2). The bedrock shows a complex fracture network and consists of alternative shale and sandstone of Cretaceous age. The weathered material was the weathering product of the sedimentary rock; both the original sedimentary structure and fracture were preserved in it without disturbance by the previous levelling. The upper part of weathered material (0–15 cm depth, WM1) was originated from shale and had a silty loam texture. The lower part (15–50 cm depth, WM2) was formed by the weathering of sandstone and had a sand texture. The beds of weathered materials and bedrock were shallowly dipping to the southeast. The attitude was 025° (counted in degrees starting clockwise) and bedding dip was 10° toward the southeast. Most joints dip 80 to 90° and opened 1 mm to 3 cm. A black-coloured coating material was observed in most macropores in WM1, WM2 and BR (Figure 4). The coating material was assumed to be reactive with conducting solute.

The characteristics of PS and overburdens (OB1 and OB2) are shown in Table I. The soils (PS, OB1 and OB2) were acidic and yellowish red to dark yellow in colour. The texture of the upper overburden (OB1) and PS was loam, and the lower overburden (OB2) had a sandy loam texture. There were three different types of structure: moderate, medium granular for OB1; moderate, medium subangular blocky structure for OB2; moderate, coarse granular structure for PS. All three soils were firm for consistency. The soils had similar

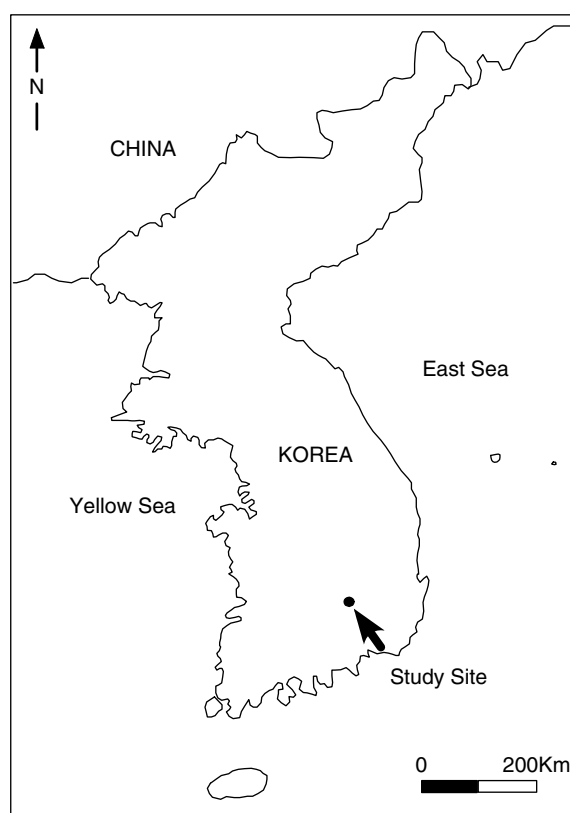


Figure 1. Location of the field study site

values of ignition loss (0.49–0.86%) as an indicator of organic matter content. Many fine rice roots were found in PS. However, plant root was not found in OB1 and OB2. The relative bulk density for the soils based on the field observation decreases in the order of OB2 > OB1 > PS. The lowest bulk density of PS is probably due to previous cultivation for rice production. Quartz, feldspars, illite and kaolinite were the major minerals in the three soils. The PS contained a trace amount of calcite.

Analysis of coating material

The black coating material in fractures that were the major conducting route of water in WM1, WM2 and BR was collected manually from the conducting fractures with a knife and the collected sample was air-dried. A portion of the air-dried coating material was finely ground with an agate mortar and pestle and analysed by X-ray diffraction (XRD; MAC Science MXP 18A Rint-2500 diffractometer) for mineral identification. The air-dried coating material that had not been ground was also analysed by scanning electron microscopy (SEM; JEOL JSL5610LV) to determine the morphology and mineralogy.

Dye tracer test

A solution of Brilliant Blue FCF (Neveon Hilton Davis Inc., Cincinnati, OH) at a concentration of 5 g l^{-1} was used to stain the flow paths (Reichenberger *et al.*, 2002). Before the application of the dye solution, the surface soil of 5 cm depth was removed and then the surface was levelled to reduce the effect of surface microrelief. An acrylic square frame ($1.0 \times 1.0 \text{ m}^2$) was installed into the ground at 5 cm depth. The framed area was flooded with 150 l of the dye solution and was covered with a vinyl film to prevent evaporation.

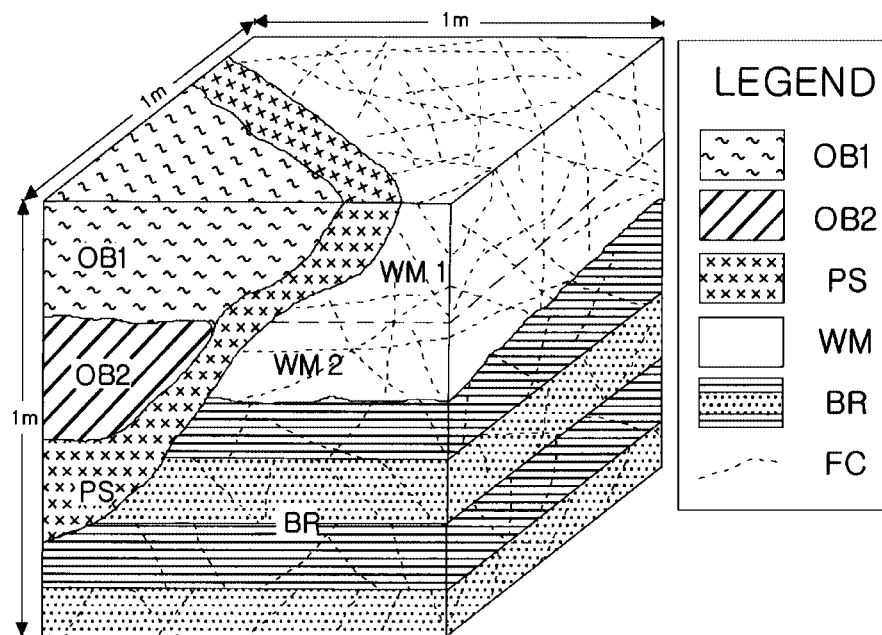


Figure 2. Schematic presentation of the profile studied. OB1: upper overburden; OB2: lower overburden; PS: buried rice paddy soil; WM1: upper weathered material; WM2: lower weathered material; BR: bedrock

Table I. Physical, chemical and mineralogical characteristics of the overburden (OB1 and OB2) and buried rice paddy soil (PS)

Soil	Colour ^a	Texture	Structure	Consistence	pH ^b	EC ($\mu\text{S}/\text{m}$) ^b	Ignition loss (%) ^c	Mineralogy ^d
OB1	10YR 5/4	Loam	Moderate, medium granular	Firm	5.70	29.8	0.49	Q, F, I, K
OB2	10YR 5/4	Sandy loam	Moderate, medium subangular blocky	Firm	4.89	25.1	0.86	Q, F, I, K, C
PS	2.5Y 4/2	Loam	Moderate, coarse granular, many fine plant roots	Firm	5.06	38.5	0.86	Q, F, I, K

^a Recorded with a soil colour reader (Minolta SPAD-503, Minolta Co. Ltd).

^b Determined with 1 : 10 soil : water method.

^c At 500 °C, overnight.

^d Determined based on XRD data. Q: quartz; F: feldspar; I: illite; K: kaolinite; C: calcite.

After 3 days of flooding, the one side of the profile was trenched down to 2 m depth for the vertical view. The horizontal views of the soil profile were exposed by removal of successive layer of 5 cm thickness using a spade from the surface down to 50 cm depth, at which the bedrock was exposed. The vertical and horizontal images were recoded using a digital camera (Sony Cybershot DSC-F717) under daylight. The same camera position was maintained for the horizontal image collection using an iron frame in order to minimize distortion differences between successive faces. A horizontal image of 80 × 80 cm² for each layer, 10 cm away from the frame, was used for image analysis to reduce the effect of the boundary flow by the acrylic frame.

Image analysis of horizontal view

All image processing was conducted using the computer image-processing program Adobe Photoshop 7.0. The pixel dimensions of the photographs were resized to 1200×1200 pixels ($80 \times 80 \text{ cm}^2$) to normalize the resolution of each photograph. Using the Color Range command of the program, the colour image taken with the digital camera was converted to a black-and-white image and the dye-stained pixels were selected and counted to calculate the dye coverage for the horizontal views.

RESULTS AND DISCUSSION

Figure 3 shows the vertical stain image of the profile. The thin surface layer ($<2 \text{ cm}$) was almost completely stained with dye. The dye penetrated down to deeper than 2 m (the maximum depth of vertical excavation for this study). A heterogeneous matrix flow was observed in OB1, WM1 and PS. Macropore flow occurred along the complex fracture network in WM1, WM2 and BR and the stained fractures had the black coating material (Figure 4). The stained fractures in WM1, WM2 and BR were concentrated in the marginal area close to the interfaces with PS. Another type of macropore flow was also observed along plant root channels in PS. Saturated interflow was observed at the continuous sloping interfaces of PS–WM1, PS–WM2 and PS–BR. The lower overburden (OB2) had no dye stain.

Figure 5 shows the black-and-white stain images of the horizontal sections at different depths. The surface was completely stained. The preferential flows along the fractures in WM1, WM2 and BR and along the continuous sloping layer interface of PS–WM1, PS–WM2 and PS–BR were the dominant flows in the profile studied. Irregular-shaped stain images suggest that heterogeneous flow was observed in the horizontal

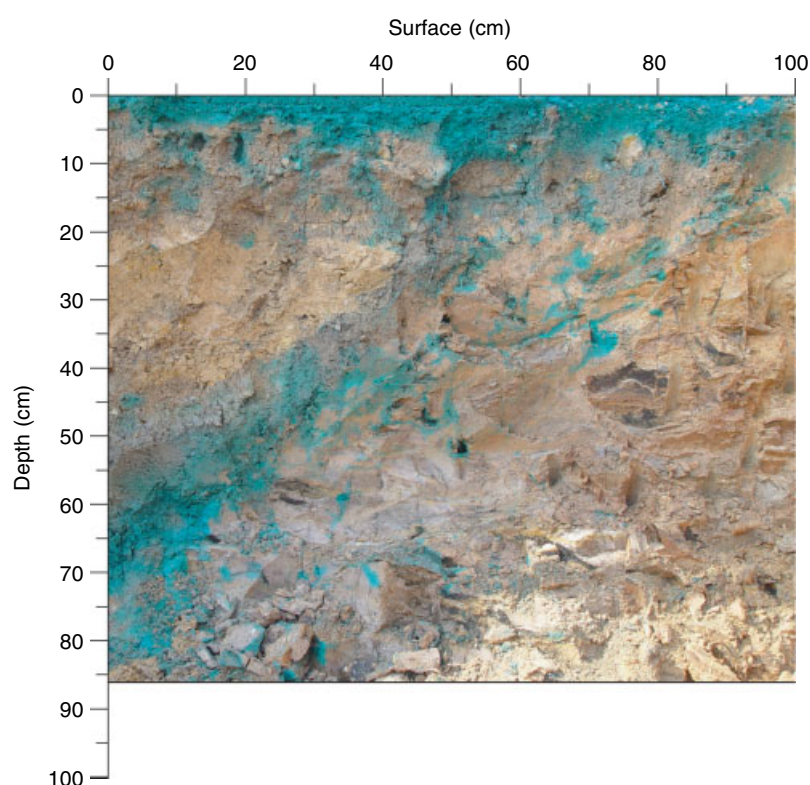


Figure 3. Stain image of the vertical cross-section of the studied profile

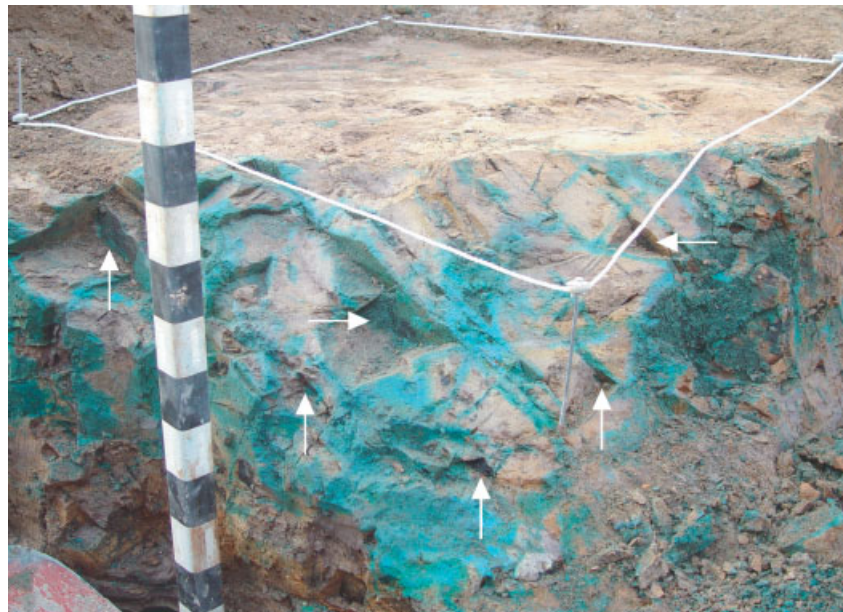


Figure 4. Selected image showing dye stain and black coating material in fracture. BC: black coating material; FC: fracture

views of PS and OB1. For weathered material, the preferential macropore flow along fractures was dominant in WM1, but a matrix flow was dominant in WM2. The preferential flow along fractures was concentrated in a narrow zone within WM1, WM2 and BR close to the sloping interface, although the fractures were rather evenly distributed in all the horizontal views of WM1, WM2 and BR. The conducting solute in weathered material infiltrated along the fractures to a shallow depth (<5 cm thickness). The calculated dye coverage with depth for the horizontal images is shown in Figure 6. Only the top portion of the profile was completely stained with the dye, and the dye coverage decreased sharply with increasing depth.

Figure 7 shows a schematic presentation of the infiltration pattern in the profile based on the interpretation of vertical and horizontal stain images. The major routes for subsurface water flow in the profile were the continuous sloping layer interface of PS–WM1, PS–WM2 and PS–BR and the fractures in weathered material and BR. The buried rice paddy soil (PS) with many fine roots and lower bulk density was another important conducting route for water flow. On the other hand, the lower overburden (OB2) acted as a ‘physical barrier’ for downward movement of infiltrating water.

The weathered material and bedrock acted as hydraulically less transmissive layers. They limited the vertical movement of water and induced saturated interflow along the sloping interface (Kung, 1990). Some of the conducting water along the interface infiltrated into the vertical and near-vertical fractures in the upper part of WM1, WM2 and BR. The cultivation of the buried rice paddy soil for a long time resulted in its low bulk density due to the formation of lots of macropores by the decay of residual rice roots. The macropores and low bulk density of PS was a cause of the formation of another major flow channel within the profile. The vertical movement of water via a heterogeneous matrix flow in OB1 paused at the layer interface between fine OB1 and coarse OB2, and thus the water flowed horizontally due to the capillary barrier (Hill and Parlange, 1972; Kung, 1990; Walter *et al.*, 2000). When the front arrived at the interface, the infiltrating water was of too high a suction to allow its entry into the pores of the coarse-textured sublayer. This resulted in no dye staining of OB2. In the WM1, WM2 and BR sequence, water moved vertically via a heterogeneous matrix flow and the preferential flow along fractures in the loosely packed upper part of WM1, and then the downward-moving water converged into the fractures in the lower part of WM1. When the downward-moving water along fractures reached the interface of WM1 and WM2 it spread out into the coarse-textured WM2,

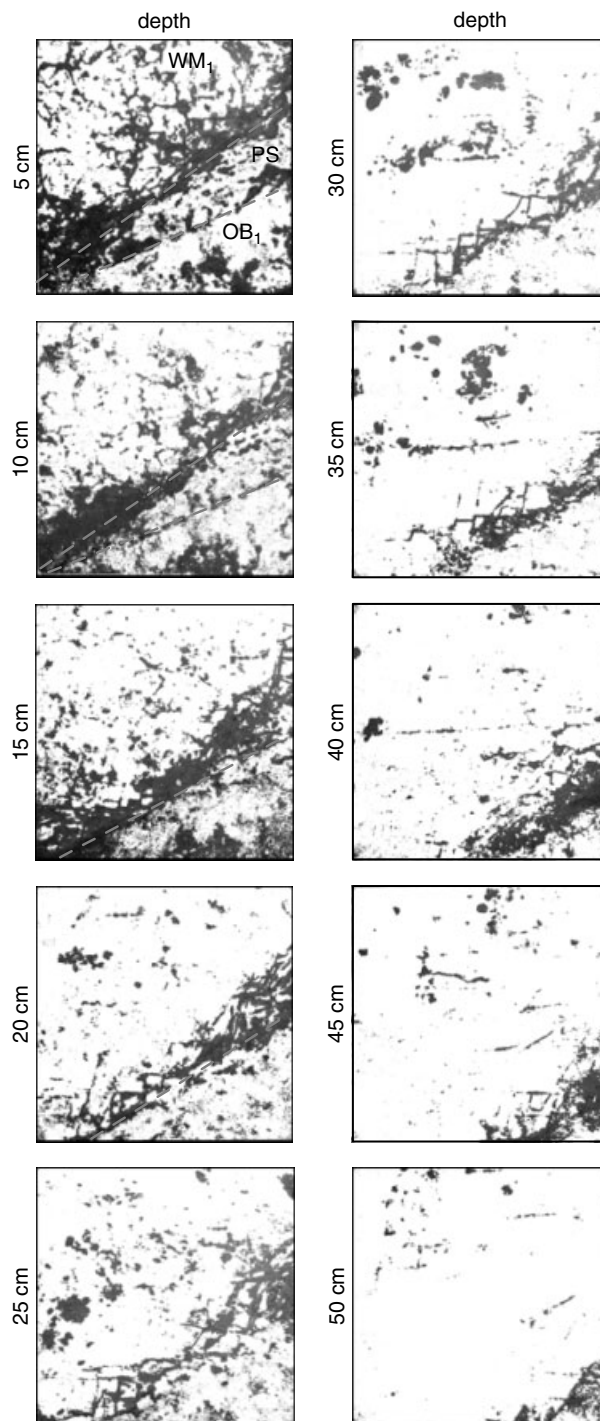


Figure 5. Black-and-white dye-stain images for the horizontal sections of the profile at different depths. OB₁: upper overburden; PS: buried rice paddy soil; WM₁: upper weathered material; dashed lines: interface

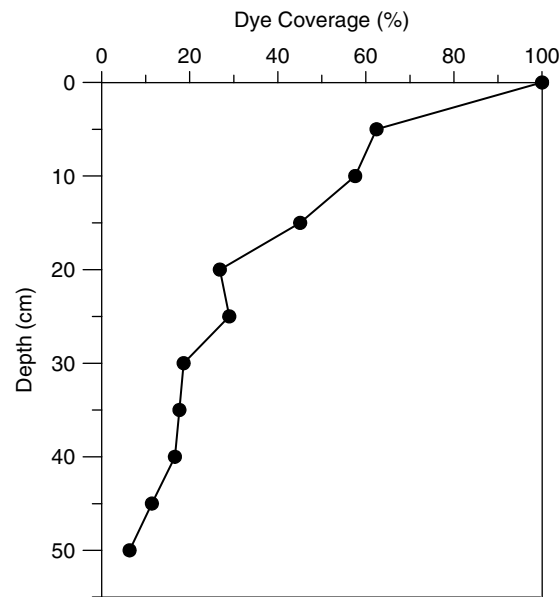


Figure 6. Calculated dye coverage with depth for horizontal stain images

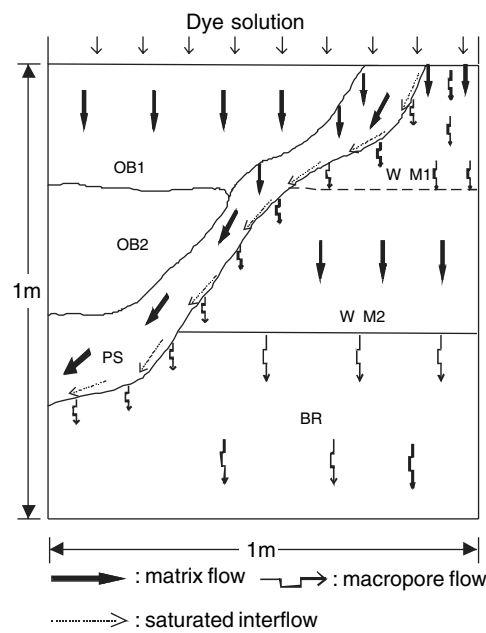


Figure 7. Schematic presentation of infiltration patterns in the profile based on the stain images of vertical and horizontal sections. OB1: upper overburden; OB2: lower overburden; PS: paddy soil; WM1: upper silty loam weathered material; WM2: sandy weathered material; BR: bedrock

forming a heterogeneous matrix flow. Finally, the water passed through WM2 and was merged into fractures in BR. The stained fractures also had a black coating material. The coating material in the stained fractures indicates that the fractures acted as conducting routes for water from the surface.

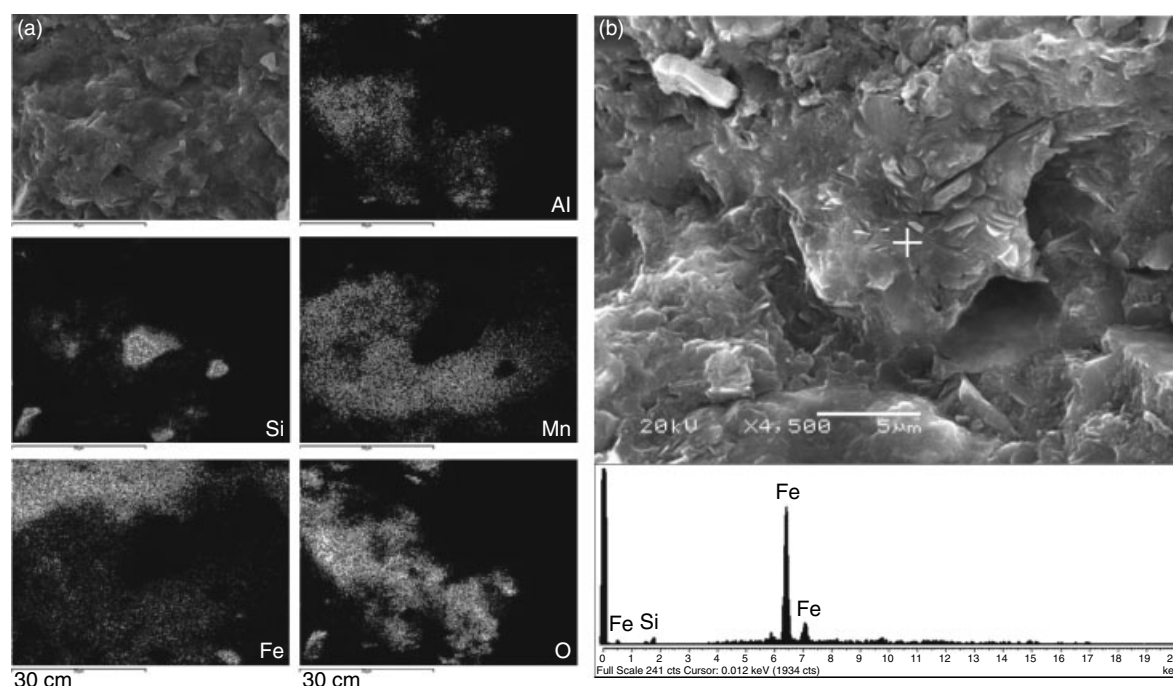


Figure 8. (a) SEM micrograph and X-ray map of the black coating material. (b) SEM micrograph and energy-dispersive X-ray pattern of iron oxide in the coating material. Haematite shows hexagonal platy morphology

The XRD data and SEM data of the fracture coating material indicates that it is a mixture of silicates and manganese and iron oxides. The SEM micrographs revealed that manganese and iron oxides occurred as clay- to silt-sized thin films coating on the silicates (Figure 8a and b). Haematite was identified as the iron oxide in the coating material. Even though a significant amount of manganese oxide in the coating material was observed under SEM, the specific mineral phase of this manganese oxide was not identified with the XRD data. This was probably due to the low crystallinity of the manganese oxide. The fine particles and the low crystallinity of manganese and iron oxides may indicate their high reactivity with contaminants.

According to previous work (McKenzie, 1989), manganese oxides have a strong oxidation potential and have a high adsorption capacity for weakly hydrated cations (such as copper, cobalt, nickel and zinc), and anions of acids (such as molybdate, phosphate and selenate). Manganese oxides increase the mobility and toxicity of chromium by the oxidation of Cr(III) to Cr(VI) in the soil and aquifer (Kim *et al.*, 2002). However, they may detoxify arsenic by oxidizing the more toxic form (arsenate) to a less toxic form (arsenite; Driehaus *et al.*, 1995). Iron oxides show a high adsorption capacity for both cationic and anionic contaminants, depending on pH (Liu and Huang, 2003; Borggaard *et al.*, 2004; Illes and Tombacz, 2004). Fine particles of manganese and iron oxides are also known as important colloidal media for the migration of contaminants in soil and aquifer (Ryan *et al.*, 1998; Swartz and Gschwend, 1998).

CONCLUSIONS

The dye-stain image clearly shows the flow paths in the regolith–bedrock profile. Both the continuous sloping layer interface of a transmissive layer over hydraulically restrictive layers (PS–WM1, PS–WM2 and PS–BR) and the fractures in WM1, WM2 and BR were the major conduits of preferential flow in the profile. The buried rice paddy soil with many fine roots and lower bulk density acted as another important water-conducting path.

The coarse-textured OB2 underlain by fine-textured OB1 acted as a physical barrier for vertical movement of water. Manganese and iron oxides in the black coating material found in the conducting fractures probably had a high surface area and low crystallinity. The properties indicate their high reactivity with conducting solute and contaminant.

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